

EE-473 - HOMEWORK 3

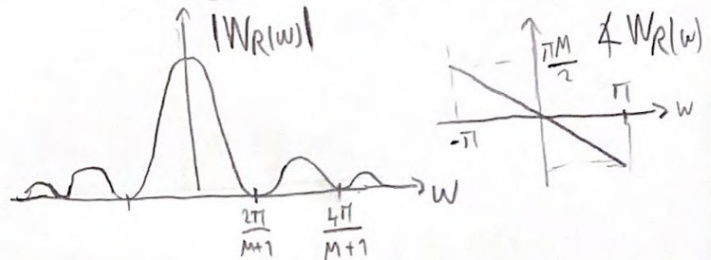
Boran Aybak Kiliç
2022401207

1)

$$a) w_R(n) = \begin{cases} 1, & 0 \leq n \leq M \\ 0, & o.w \end{cases}$$

$$W_R(w) = \sum_{n=0}^M e^{-jwn} = \frac{1 - e^{-jw(M+1)}}{1 - e^{-jw}} = \frac{e^{-jw \frac{M+1}{2}}}{e^{-jw/2}} \cdot \frac{e^{jw \frac{M+1}{2}} - e^{-jw \frac{M+1}{2}}}{e^{jw/2} - e^{-jw/2}}$$

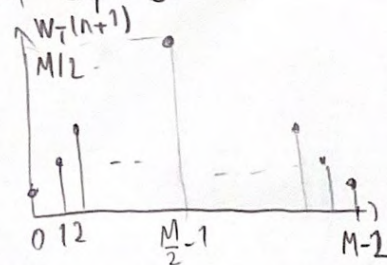
$$W_R(w) = e^{-jw \frac{M}{2}} \frac{\sin(w \frac{M+1}{2})}{\sin(w/2)}$$



$$b) w_T(n) = \begin{cases} n, & 0 \leq n \leq M/2 \\ M-n, & M/2 < n \leq M \\ 0, & o.w \end{cases}$$

↳ We can't really express this as the convolution of 2 rectangular signals with itself. HOWEVER we can express

$$w_T(n+1) = \begin{cases} n+1, & 0 \leq n \leq M/2-1 \\ M-n-1, & M/2 \leq n \leq M-1 \\ 0, & o.w \end{cases}$$



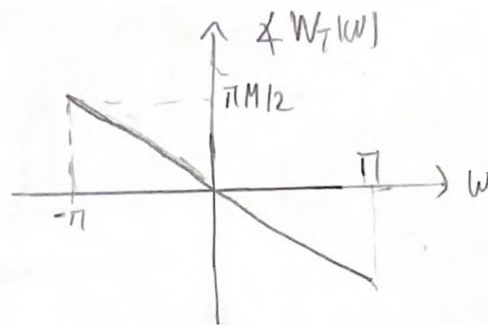
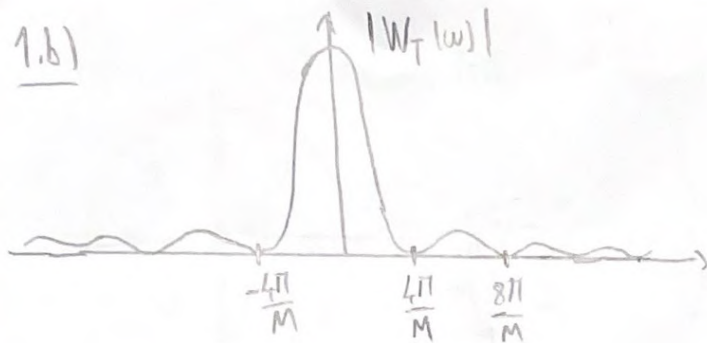
as the convolution of $w_T(n+1) = v_T(n) * v_T(n)$ where

$$v_T(n) = \begin{cases} 1, & 0 \leq n \leq \frac{M}{2}-1 \\ 0, & o.w \end{cases}$$

$$\text{Then } W_T(w) = V_T(w) e^{-jw} = e^{-jw} \left[e^{-jw \frac{M-1}{2}} \frac{\sin(\frac{wM}{4})}{\sin(w/2)} \right]^2$$

$$W_T(w) = \left(\frac{\sin(\frac{wM}{4})}{\sin(w/2)} \right)^2 e^{-jw \frac{M}{2}}$$

1.b)

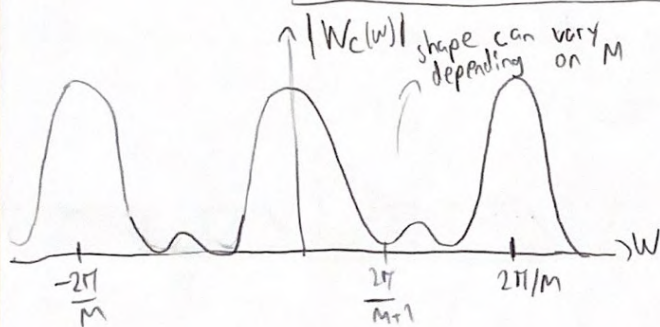


$$c) w_c(n) = \frac{1}{2} \left(1 + \cos \frac{2\pi n}{M} \right) w_R(n)$$

$$W_c(w) = \frac{1}{2\pi} \left[2\pi \delta(w) + \pi \delta\left(w - \frac{2\pi}{M}\right) + \pi \delta\left(w + \frac{2\pi}{M}\right) \right] * W_R(w)$$

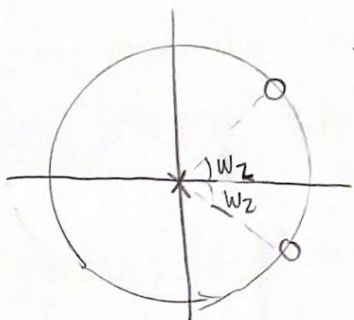
$$= \delta(w) * W_R(w) + \frac{1}{2} \delta\left(w - \frac{2\pi}{M}\right) * W_R(w) + \frac{1}{2} \delta\left(w + \frac{2\pi}{M}\right) * W_R(w)$$

$$= W_R(w) + \frac{1}{2} W_R\left(w - \frac{2\pi}{M}\right) + \frac{1}{2} W_R\left(w + \frac{2\pi}{M}\right)$$

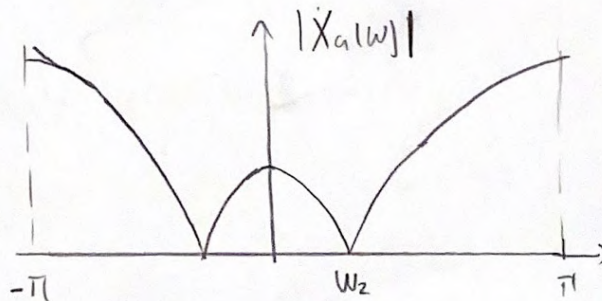


2)

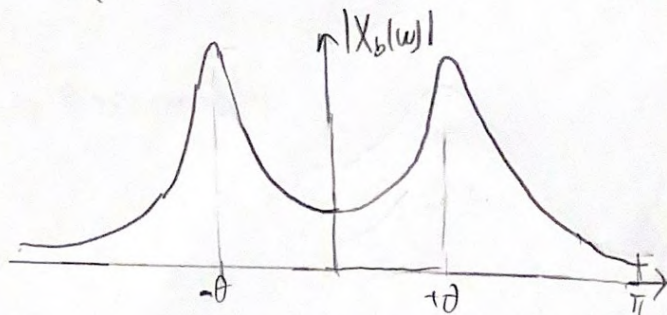
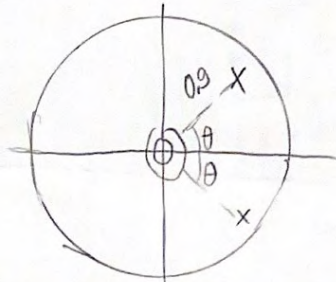
a)



→ Magnitude is zero at $w=w_2$

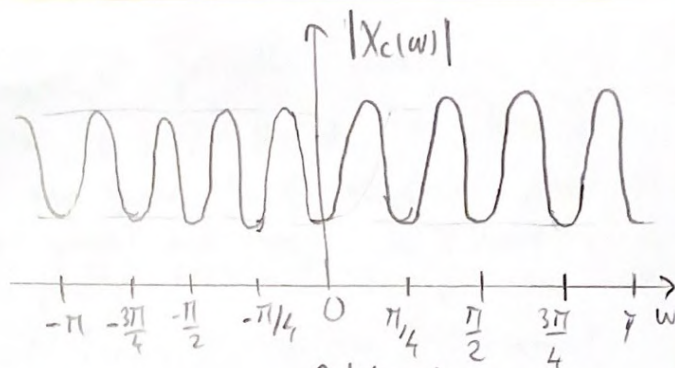


b)

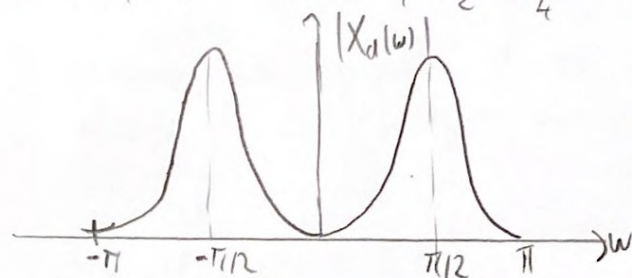
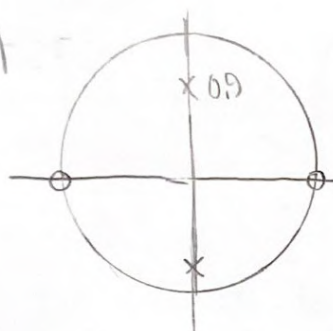


Ex 3.2c

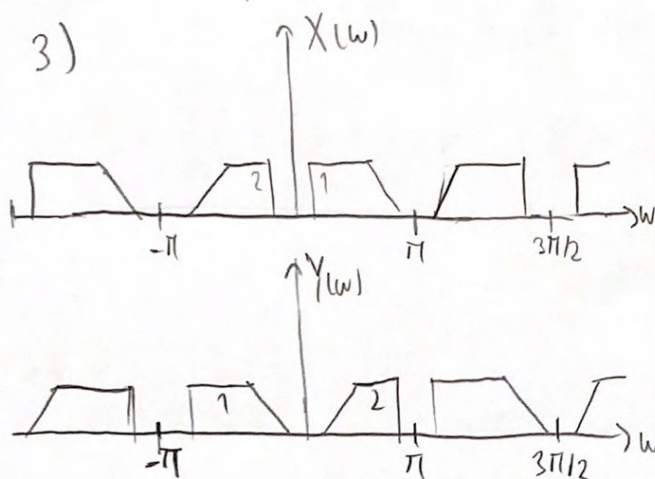
→ Comb Filter



d)



3)



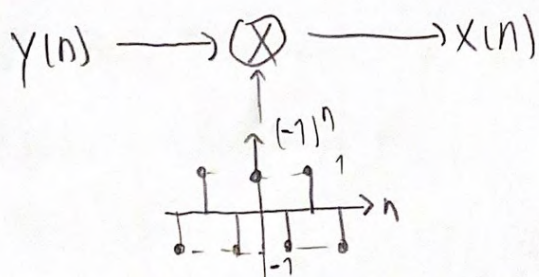
→ Notice that spectrums of discrete time signals are periodic (with 2π) and $Y(w) = X(w - \pi)$

$$a) y(n) = e^{-jw\pi} x(n)$$

$$y(n) = (-1)^n x(n)$$

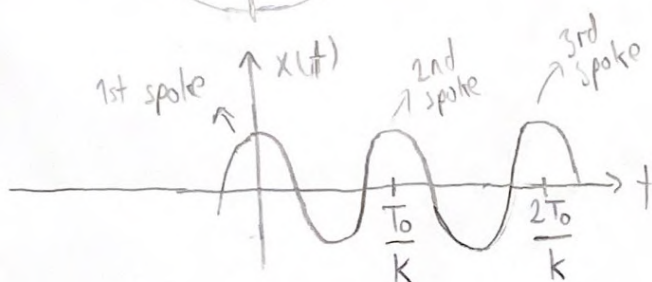
b) To unscramble we can just multiply $y(n)$ with $(-1)^n$

$$\hookrightarrow y(n) (-1)^n = x(n)$$



⇒ Unscrambler

4)



⇒ Assign a periodic function $x(t)$ which is proportional to closeness of any spoke to the downmost point of the wheel and choose a sinusoidal for simplicity. $x(t)$ will be periodic with

$T_0/k = \frac{1}{f_0 k}$ meaning that wheel becomes undistinguishable if we observe it at integer multiples of $\frac{1}{f_0 k}$.

$$\hookrightarrow x(t) = \cos\left(\frac{2\pi k}{T_0} t\right)$$

a) For wheel to appear still, we must observe it on integer multiples of $T_{\text{spoke}} = \frac{m}{f_0 k}$ ($m \in \mathbb{Z}^+$)

$$T = \frac{1}{F_s} = \frac{m}{f_0 k}$$

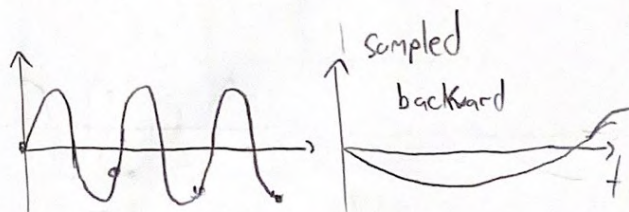
or

$$F_s = \frac{f_0 k}{m}; m=1,2,3,\dots$$

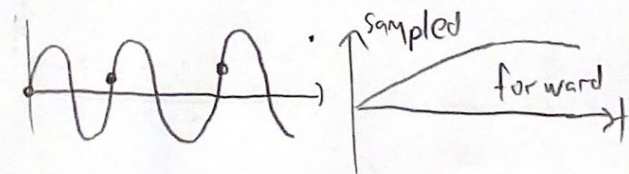
↳ Condition on frame rate (F_s) for wheels to appear stand still.

b)

If $k \cdot F_s > F_{\text{spoke}} = \frac{k}{T_0}$ wheels appear to move backward ($k \in 1,2,3,\dots$)



If $k \cdot F_s < F_{\text{spoke}} = \frac{k}{T_0}$ wheels appear to move forwards.



$$v = 25 \text{ km/hr} \approx 6.944 \text{ m/s} \Rightarrow f_0 = \frac{v}{\pi D} = \frac{6.944 \text{ m/s}}{\pi \times 1.5} \approx 1.474 \text{ Hz}$$

$$F_{\text{spoke}} = k f_0 \approx 58.94 \text{ Hz}$$

→ We choose $k=4$ (since it makes $k F_s$ closest to 58.94 Hz)

$$\left. \begin{array}{l} k F_s = 60 \text{ Hz} \\ F_{\text{spoke}} = 58.94 \text{ Hz} \end{array} \right\} \begin{array}{l} \text{Since } k F_s \text{ is slightly higher than } F_{\text{spoke}}, \\ \text{wheel will appear to spin backwards} \end{array} \quad \begin{array}{l} f_{\text{alias}} \\ -1.06 \text{ Hz} \end{array}$$

$$5) \quad x_a(t) = \begin{cases} e^{-j2\pi F_0 t} & , t \geq 0 \\ 0 & , t < 0 \end{cases}$$

$$a) \quad X_a(F) = \int_{-\infty}^{\infty} x_a(t) e^{-j2\pi Ft} dt = \int_0^{\infty} e^{-j2\pi(F+F_0)t} dt = \boxed{\frac{1}{j2\pi(F_0+F)}}$$

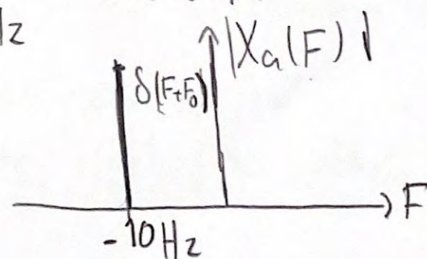
$$b) \quad x(n) = x_a(nT) = \begin{cases} e^{-j2\pi F_0 T n} & , n \geq 0 \\ 0 & , n < 0 \end{cases}$$

$$X(f) = \sum_{n=-\infty}^{\infty} x(n) e^{-j2\pi f n} = \sum_{n=0}^{\infty} e^{-j2\pi(f+F_0 T)n} = \boxed{\frac{1}{1 - e^{-j2\pi(f + \frac{F_0}{F_s})}}}$$

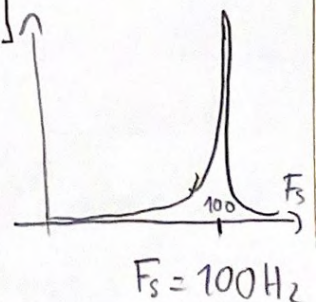
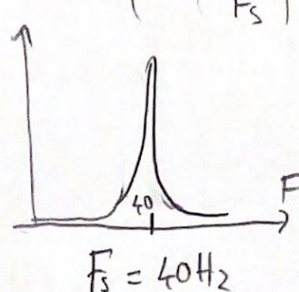
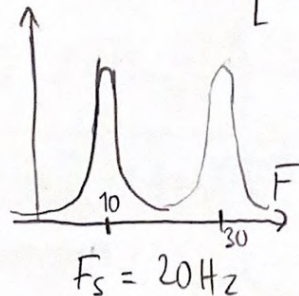
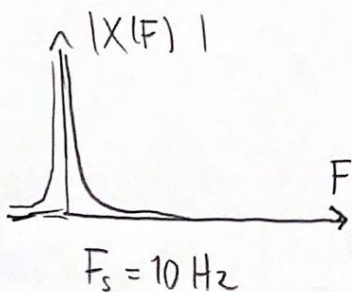
$$c) \text{ Magnitude spectrum of } X_a(F): |X_a(F)| = \frac{1}{2\pi(F_0+F)}$$

↳ This has a infinity at $F = -10\text{Hz}$

↳ Effectively, this dirac at -10Hz will be all that can be seen.



$$d) \quad |X(F)| = \frac{1}{|1 - e^{-j2\pi(F+F_0)/F_s}|} = \frac{1}{[2 - 2\cos(2\pi|\frac{F+F_0}{F_s}|)]^{1/2}}$$



e) In $F_s = 10\text{Hz}$, baseband is distorted and signal will be aliased

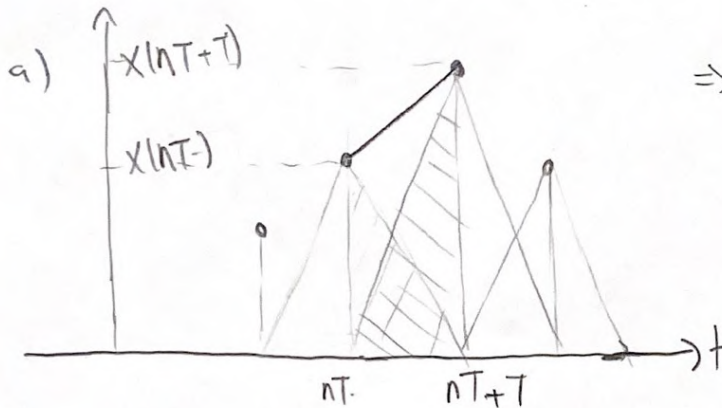
↳ Since under-sampled $k=1$ copy of dirac at $F = -10\text{Hz}$ will create a huge DC component which shouldn't be there

→ In $F_s = 20\text{Hz}$, sampled signal is right at Nyquist boundary. Rest are not aliased.

$$6) \hat{x}(t) = x(nT - T) + \frac{x(nT) - x(nT - T)}{T} (t - nT), \quad nT \leq t \leq (n+1)T$$

↳ Linear interpolation: connecting successive points by line segments.

$$\Rightarrow \left. \begin{array}{l} \text{At } t = nT \quad \hat{x}(t) = x((n-1)T) \\ \text{At } t = nT + T \quad \hat{x}(t) = x(nT) \end{array} \right\} \begin{array}{l} \text{inherent delay of } T \text{ seconds} \\ \text{in this interpolation technique} \end{array}$$



$$\Rightarrow \text{Let } x(t) = \sum_{k=-\infty}^{\infty} x(kT) \delta(t - kT)$$

$$\hat{x}(t) = x(t) * h(t)$$

$$= \sum_{k=-\infty}^{\infty} x(kT) h(t - kT)$$

↳ Sum of shifted and scaled interpolation kernels

→ We can first find the linear interpolation with no delay and shift the kernel by T to the right, to obtain the T -second delay linear interp. kernel

$$\hat{x}_{\text{no-delay}} = x(nT) + \frac{x(nT+T) - x(nT)}{T} (t - nT)$$

$$= x(nT) \underbrace{\left(1 - \frac{t - nT}{T}\right)}_{h_{\text{right}}(t - nT)} + x(nT+T) \underbrace{\left(\frac{t - nT}{T}\right)}_{h_{\text{left}}(t - (n+1)T)}$$

$$\rightarrow h_{\text{right}}(t) = \begin{cases} h(t), & 0 \leq t < T \\ 0, & \text{o.w.} \end{cases} \quad \rightarrow h_{\text{left}}(t) = \begin{cases} h(t), & -T \leq t < 0 \\ 0, & \text{o.w.} \end{cases}$$

$$\left. \begin{array}{l} h_{\text{right}}(t) = 1 - \frac{t}{T} \\ h_{\text{left}}(t) = 1 + \frac{t}{T} \end{array} \right\} h_{\text{no-delay}}(t) = \begin{cases} 1 - \frac{t}{T}, & 0 \leq t < T \\ 1 + \frac{t}{T}, & -T \leq t < 0 \\ 0, & \text{o.w.} \end{cases}$$

$$\text{Then } h(t) = h_{\text{no-delay}}(t-T) = \begin{cases} +1/T & , 0 \leq t < T \\ 2-t/T & , T \leq t < 2T \\ 0, w & , o.w \end{cases}$$

$$H(F) = \int_{-\infty}^{\infty} h(t) e^{-j2\pi Ft} dt = \underbrace{\frac{1}{T} \int_0^T 1 e^{-j2\pi Ft} dt}_{I_1} + \underbrace{2 \int_T^{2T} (2-t/T) e^{-j2\pi Ft} dt}_{I_2} + \underbrace{-\frac{1}{T} \int_T^{2T} 1 e^{-j2\pi Ft} dt}_{I_3}$$

$$\text{Let } a = -j2\pi f \text{ and } I_1 = \frac{1}{T} \int_0^T 1 e^{+at} dt, I_2 = 2 \int_T^{2T} e^{+at} dt, I_3 = \frac{1}{T} \int_T^{2T} 1 e^{+at} dt$$

$$I_1 = \frac{1}{T} \left(\frac{T e^{aT}}{a} - \frac{e^{aT} - 1}{a^2} \right) = \frac{e^{aT}}{a} - \frac{e^{aT}}{a^2 T} + \frac{1}{a^2 T}$$

$$I_2 = \frac{2 e^{a2T}}{a} - \frac{2 e^{aT}}{a}, \quad I_3 = -\frac{1}{T} \left(\frac{2T e^{2aT} - T e^{aT}}{a} - \frac{e^{2aT} - e^{aT}}{a^2} \right)$$

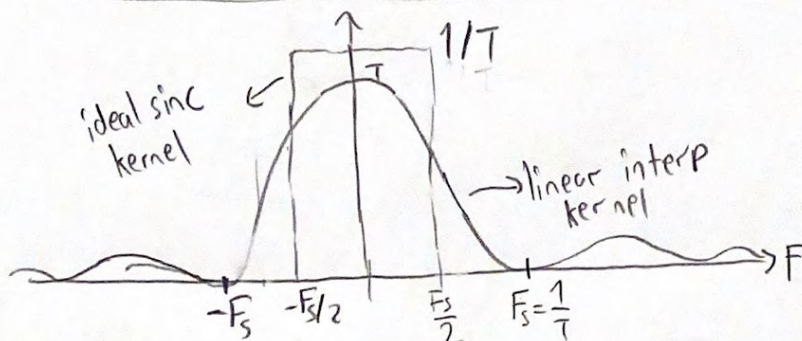
$$= -\frac{2}{a} e^{2aT} + \frac{e^{aT}}{a} + \frac{e^{2aT}}{a^2 T} - \frac{e^{aT}}{a^2 T}$$

$$H(F) = \frac{1}{a^2 T} \left[1 - e^{aT} - e^{aT} (1 - e^{+aT}) \right] = \frac{(1 - e^{+aT})^2}{a^2 T}$$

$$= \frac{(e^{+aT/2})^2}{a^2 T} \left(\frac{e^{-aT/2} - e^{+aT/2}}{a} \right)^2 = \frac{e^{-j2\pi f T}}{T} \left(\frac{e^{j\pi f T} - e^{-j\pi f T}}{2j\pi f} \right)^2$$

$$H(F) = \frac{e^{-j2\pi f T}}{T} \left(\frac{\sin(\pi f T)}{\pi f} \right)^2$$

b)



→ Obviously linear interp includes freq. components outside of base band producing an inaccurate interp